Lecture 3 – Economic Evaluation

Economics of Health and Health Care
Spring 2026 S2-T3

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A (true) story

- June 2009 – UK – Cancer patients wait a decision on a new life-extending drug
- Drug *Revlimid* treats multiple myeloma – should it be prescribed freely?
- If not included in the NHS coverage – too expensive for people to pay
- NICE (official institution) decides based on **QALY – Quality Adjusted Life Years** and **ICER – Incremental Cost Effectiveness Ratio**

Economic assessment

- ICER (the lower the better):
  GBP 24,584 per QALY
- Cut-off: GBP 30,000
- NICE stated Celgene's model was incomplete
- Also corrected a technical point – using the mean survival rate and not the median
- ICER changes to GBP 47,100: now rejected, above the threshold for value

- Revision of the model by Celgene implied patients could expect to live 1,8 years longer, at a cost of 54,291 GBP per patient; 1,8 years amount to 1,24 QALYs
- Quality of life of revmilid patients is rated 0,68
- ICER is now GBP 43,800
- Should it be funded?

What if the money has to come from

- Babies support in areas with high infant mortality
- No decent palliative care for other terminally ill patients
- No new diagnostic equipment for a hospital

What do these terms (**ICER**, **cost-effectiveness**) really mean?

Economic evaluation

- Cost minimization
- Cost – effectiveness
- Cost – utility
- Cost – benefit

Measuring benefits: utility => QALY

Economic evaluation

- Steps to take:
  - Question addressed – viewpoint + alternatives
  - Define and assess benefits and costs, define type of analysis
  - Take care of timing/discounting
  - Take care of uncertainty
  - Take care of medical decision making if appropriate (e.g. Markov chains)

Measuring benefits: QALY

- QALY: Quality-Adjusted Life Years
- This concept is one of the most important contributions of economic theory to economic evaluation analysis in the health economy.
- It attempts to overcome the limitations imposed in clinical measures like *survival rate*
- It introduces two dimensions: longevity and quality of life.

QALY

- QALYs rely on two refutable hypothesis:
  - That the health benefit for each individual is equal to the sum of health gains (quality of life) over the years the individual benefits from the effect of medical care
  - That the value to society is equal to the sum of health benefits produced

QALY

QALY - each remaining year of a person's life is weighted by the quality of life expected in each future year.

Example:

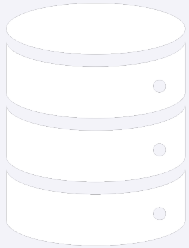
A 70 year old man who expects to live for another 20 years.	Up to 80 years old, he expects to have no health problems;	From 80 to 90 years old, he expects to have half the quality of life of the previous 10 years.	QALYs = 10 x 1 + 10 x 0.5 = 15

QALY

The value of attaining a certain health state to an individual is the product of two factors:

The increase in utility associated with the person's health status

The number of years this improvement is expected to hold.



QALY is commonly referred to as a utility index.

QALY

This “utility” can be measured by:		
“standard gamble”	“time trade-off” (TTO)	“ratio scale”

QALYs standard gamble

The procedure to use “standard gamble” is the following:

- Two alternatives are offered.
  - Alternative A has two possible outcomes:
    - Return to the state of perfect health for t years (with probability p)
    - Immediate death (with probability 1-p)
  - Alternative B is to maintain a given condition for t years.
- Then it is asked what probability *p* would make the individual indifferent between the two alternatives.

That probability is the utility index we are looking for!

QALYs time trade-off

The procedure to use “time trade-off” is the following:

- Two alternatives are offered
- In alternative A, the individual will have a health status of H for t years, and die right after that;
- In alternative B, a perfect health status is attained for x years, followed by death.
- What is the value of x that makes equivalent the two alternatives?

The ratio x/t is the score we are looking for!

Example

- Suppose someone is in bed for life, and has a life expectancy of 30 years more
- Ask how many years in perfect health are equivalent to the current situation?

$$V(in\text{-}bed)x30=kV(healthy)$$

- Suppose k=10.
- Assumption (at no cost – V(healthy)=1)
- Then V(in-bed)=10/30=1/3.

QALYs ratio scaling

When using ratio scaling, values result from asking people directly how they value each health state on a linear scale, for example, from 0 to 100,

This values are then converted to the [0,1] interval

QALYs

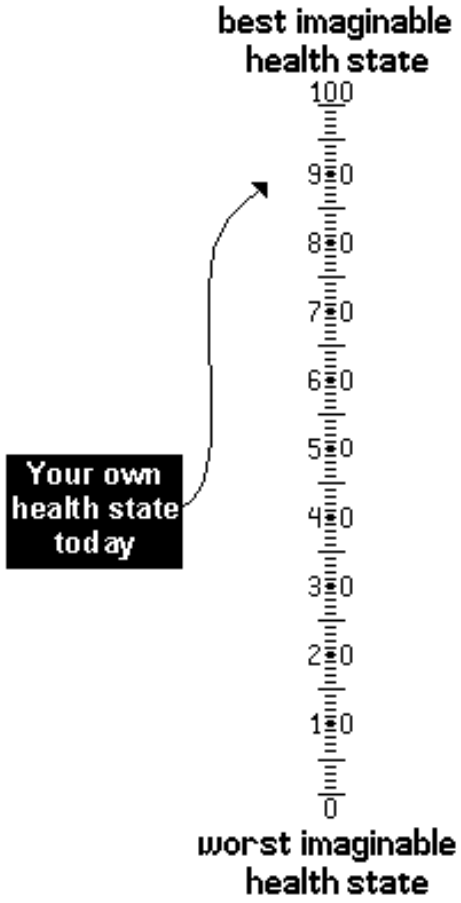
It is not mandatory to build the utility index from scratch in order to use QALYs

Existing instruments like EuroQoL (EQ-5D) may be used

An example - EuroQoL

- Each state has 5 attributes
- Each attribute has three levels - 243 possible health states, plus “unconscious” and “dead” – adding to a total of 245 health states to quantify
- Quantification based on Time Trade-Off -United Kingdom
- Current version has been extended to 5 levels – EQ-5D-5L

EuroQoL – Classification system



- MOBILITY
  - **I have no problems in walking about**
  - **I have some problems in walking about**
  - **I am confined to bed**
- SELF-CARE
  - **I have no problems with self-care**
  - **I have some problems washing or dressing myself**
  - **I am unable to wash or dress myself**
- USUAL ACTIVITIES (e.g. work, study, housework family or leisure activities)
  - **I have no problems with performing my usual activities**
  - **I have some problems with performing my usual activities**
  - **I am unable to perform my usual activities**
- PAIN/DISCOMFORT
  - **I have no pain or discomfort**
  - **I have moderate pain or discomfort**
  - **I have extreme pain or discomfort**
- ANXIETY/DEPRESSION
  - **I am not anxious or depressed**
  - **I am moderately anxious or depressed**
  - **I am extremely anxious or depressed**

EuroQoL Scoring formula – UK tariffs

Coefficients for TTO tariffs

DIMENSION	COEFFICIENT
Constant	0.081
Mobility	
level 2	0.069
level 3	0.314
Self-care	
level 2	0.104
level 3	0.214
Usual activity	
level 2	0.036
level 3	0.094
Pain/discomfort	
level 2	0.123
level 3	0.386
Anxiety/depression	
level 2	0.071
level 3	0.236
N3	0.269

Observations

- Scores for any given state are calculated subtracting the relevant coefficient from 1 (which represents perfect health)
- If there is any dysfunction at all – that is, level > 1 in any dimension - subtract the constant term
- If there is at least one level 3 in any dimension, subtract the other constant term (N3)

Example 11223(Drummond et al. pp.164)

Full health	=1.000
Constant term (for any dysfunctional state)	−0.081
Mobilty (level 1)	−0
Self-care (level 1)	−0
Usual activities (level 2)	−0.036
Pain or discomfort (level 2)	−0.123
Anxiety or depression (level 3)	−0.236
N3 (level 3 occurs within at least one dimension)	−0.269
Therefore, the estimated value for 11223	=0.255

	eq5d_GB	eq5d_ES	eq5d_FR	eq5d_IT	eq5d_NL
1.	.883	.905	.844	.915	.897
2.	1	1	1	1	1
3.	1	1	1	1	1
4.	1	1	1	1	1
5.	.848	.914	.91	.905	.805
6.	.812	.8430001	.754	.863	.773
7.	.796	.887	.888	.902	.843
8.	.689	.7540001	.642	.808	.687
9.	1	1	1	1	1
10.	1	1	1	1	1
11.	1	1	1	1	1
12.	.796	.887	.888	.902	.843
13.	.848	.914	.91	.905	.805
14.	1	1	1	1	1
15.	1	1	1	1	1
16.	1	1	1	1	1
17.	.848	.914	.91	.905	.805
18.	.848	.914	.91	.905	.805

- We now have these “tariffs” for many countries
- Including Portugal since 2013 !

	UK	Spain	Portugal
	Coef	Coef	Coef
Constant	0,081	0,024	
Mobility			
Level 2	0,069	0,106	0,305
Level 3	0,314	0,430	0,890
Self-care			
Level 2	0,104	0,134	0,354
Level 3	0,214	0,309	0,796
Usual Activity			
Level 2	0,036	0,071	0,298
Level 3	0,094	0,195	0,759
Pain			
Level 2	0,123	0,089	0,306
Level 3	0,386	0,261	0,703
Anxiety			
Level 2	0,071	0,062	0,233
Level 3	0,236	0,144	0,632
N3	0,269	0,291	
D1			-0,437
I2			0,240
$I3^2$			-0,034

The new EQ-5D-5L

- Becomes finer in the number of levels per dimension
- Requires more effort in finding the ”tariffs” for each level

Health Questionnaire (EQ-5D-5L)

Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY

☐1 I have no problems in walking about
☐2 I have slight problems in walking about
☐3 I have moderate problems in walking about
☐4 I have severe problems in walking about
☐5 I am unable to walk about

SELF-CARE

☐1 I have no problems washing or dressing myself
☐2 I have slight problems washing or dressing myself
☐3 I have moderate problems washing or dressing myself
☐4 I have severe problems washing or dressing myself
☐5 I am unable to wash or dress myself

USUAL ACTIVITIES *(e.g. work, study, housework, family or leisure activities)*

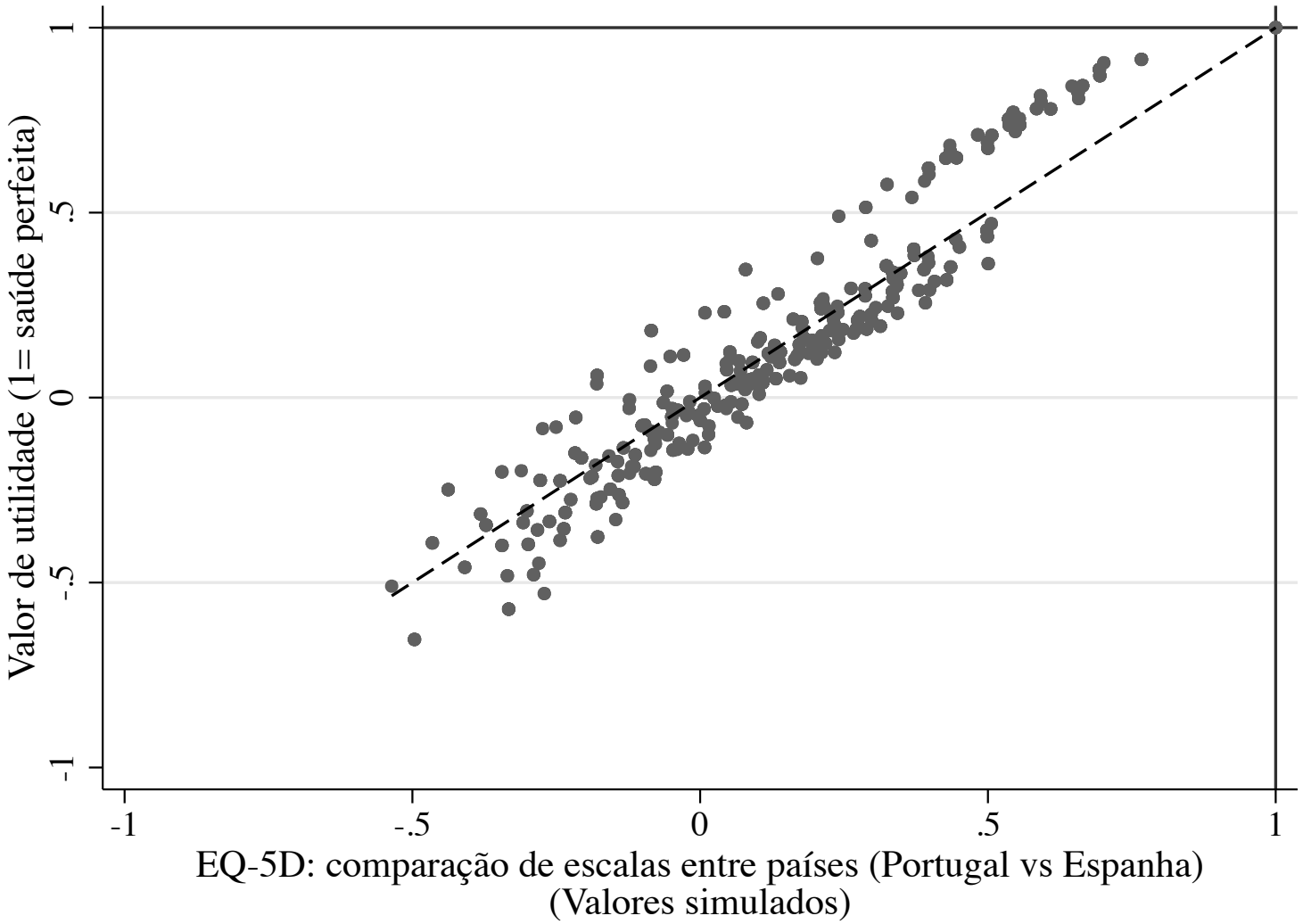
☐1 I have no problems doing my usual activities
☐2 I have slight problems doing my usual activities
☐3 I have moderate problems doing my usual activities
☐4 I have severe problems doing my usual activities
☐5 I am unable to do my usual activities

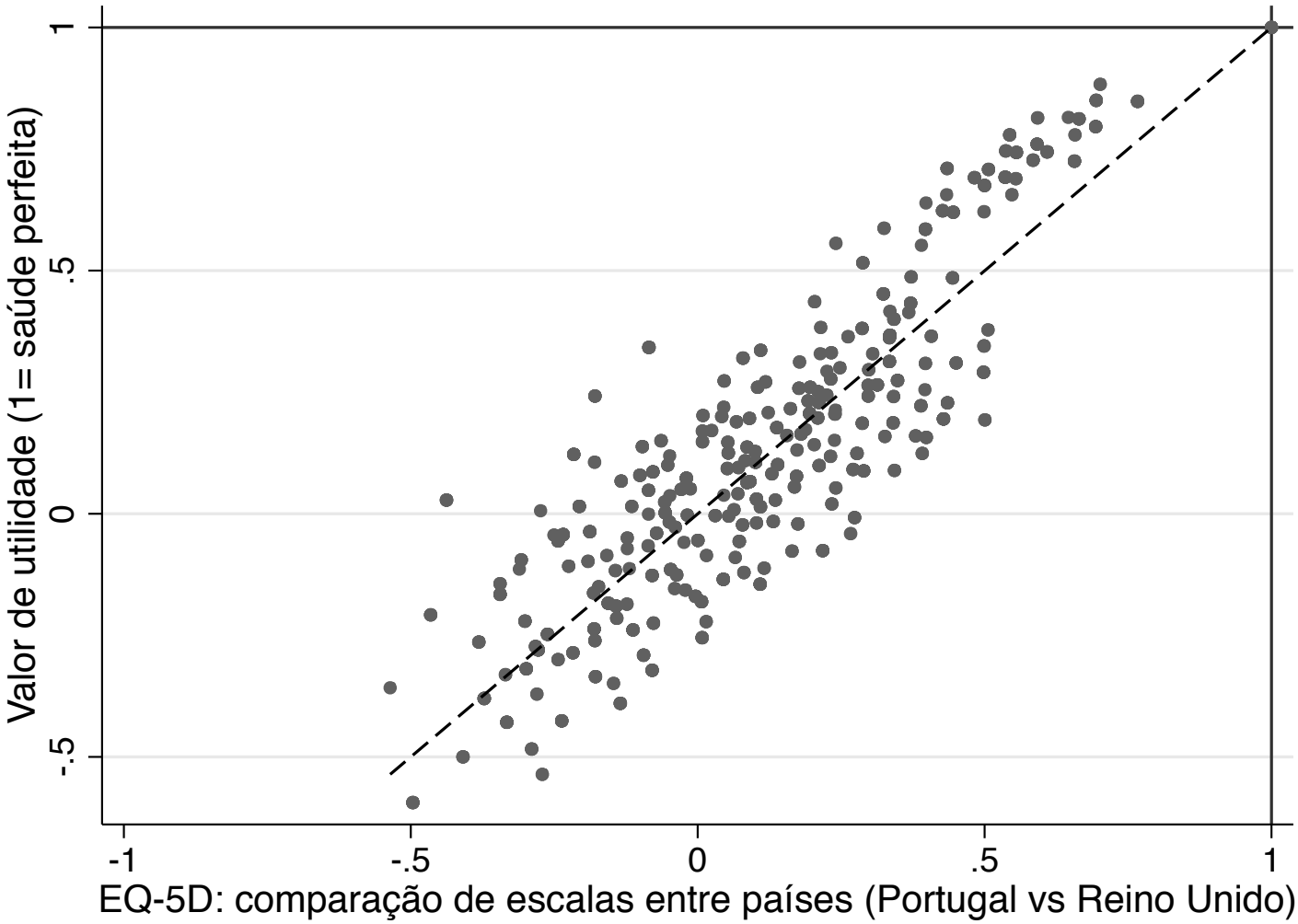
PAIN / DISCOMFORT

☐1 I have no pain or discomfort
☐2 I have slight pain or discomfort
☐3 I have moderate pain or discomfort
☐4 I have severe pain or discomfort
☐5 I have extreme pain or discomfort

ANXIETY / DEPRESSION

☐1 I am not anxious or depressed
☐2 I am slightly anxious or depressed
☐3 I am moderately anxious or depressed
☐4 I am severely anxious or depressed
☐5 I am extremely anxious or depressed


Valor de utilidade (1= saúde perfeita)
1
.5
0
-.5
-1
-1
-.5
0
.5
1
EQ-5D: comparação de escalas entre países (Portugal vs Espanha)
(Valores simulados)


Valor de utilidade (1= saúde perfeita)
1
.5
0
-.5
-1
-.5
0
.5
1
EQ-5D: comparação de escalas entre países (Portugal vs Reino Unido)

Costs

Costs are easier to treat

Usually specified in monetary terms

The usual concepts apply: fixed and marginal cost, average cost, incremental cost

Direct and indirect costs (costs to family and friends; production lost)

Valuations – market prices or “best proxy” for opportunity costs

Criterion

- Criterion: benefits – costs
- But requires they are measured in the same unit
- More often: ICER – Incremental Cost-Effectiveness Ratio: (Ct – Ca)/(Bt – Ba)
- Extra gains compensate extra costs?
- (makes very important the comparison point)
- Relates to decision under budget constraint